Day 12 8/11/26

876. Middle of the Linked List

Approach: Slow and Fast Points
best approach is to use two pointers:

- Slow moves one step at a time.
- Fast moves two steps at a time.

When fast reaches the end, slow will be the middle of the linked list.

Code:-

```
class Solution {
    public ListNode middleNode(ListNode head) {
        ListNode slow = head;
        ListNode fast = head;

        while (fast != null && fast.next != null) {
            slow = slow.next;
            fast = fast.next.next;
        }
        return slow;
    }
}
```

Time Complexity: O(n)
Space Complexity: O(1)

Description | Accepted | Editorial | Solutions | Submissions

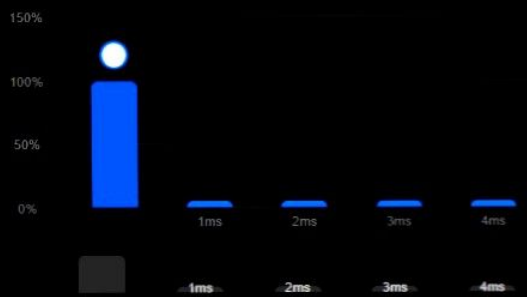
All Submissions

Runtime

0 ms | Beats **100.00%**

Memory

42.64 MB | Beats 62.96%



Code Java

```
1 /**
```

Code

Java | Auto

```
11  class Solution {
12      public ListNode middleNode(ListNode head) {
13
14          ListNode slow = head , fast = head;
15
16          while(fast != null && fast.next != null){
17              slow = slow.next;
18              fast = fast.next.next;
19          }
20          return slow;
21      }
```

Saved

Ln 1, Col 1

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

head =

[1,2,3,4,5]

Output